Chayma EL HOUMA

Backend Java Developer | Spring Boot

Email chaymaelhouma2005@gmail.com

Phone +212 6 72 98 45 69

Portfolio chaymae-portfolio.vercel.app

LinkedIn linkedin.com/in/chaymaelhouma

GitHub github.com/ChaymaELHOUMA

Profile

Final-year Computer Science and Artificial Intelligence engineering student specializing in Java/Spring Boot backend development. Experienced in designing RESTful APIs, scalable backend systems, and integrating Generative AI technologies such as LLMs and Retrieval-Augmented Generation into software solutions. Seeking a Final-Year Engineering Internship (PFE) to contribute to enterprise-grade Java applications.

Education

Engineering Degree Program – Computer Science and Artificial Intelligence

ENSA Safi | 2024 – Present

Integrated Preparatory Cycle

ENSA Safi | 2022 – 2024

Baccalaureate in Physical Sciences

Lycée Awlade Amrane, Sidi Bennour | 2021 – 2022 | High Honors

Professional Experience

Backend Java Developer Intern – CyberWatch360 Project

Arimayi – France (Remote) | June 9, 2026 – Present

- Analyzed functional requirements and designed UML diagrams for Module 2 (Asset Management) of the CyberWatch360 cybersecurity platform.
- Developed five production-ready backend user stories in Java/Spring Boot for Module 2: asset management, import, editing, multi-criteria search, and dependency management between assets.
- Designed wireframes in Figma, created Postman API collections, implemented unit tests, authored technical documentation, and collaborated using Git Flow.

Technologies: Java 17, Spring Boot, Spring Modulith, Spring Security, Spring Data JPA, PostgreSQL, Flyway, MapStruct, Lombok, Docker, Maven, JUnit, Mockito, Git, GitHub, Postman, Figma

Software Test Automation Intern

Smart Automation Technologies – Tangier, Morocco (Remote) | July 9, 2026 – September 9, 2026 (Expected)

- Developing a Java 17/Spring Boot backend orchestrator integrating Large Language Models (LLMs) for automated software test generation.
- Designing REST APIs and LLM prompt construction from AST-based code analysis to generate and evaluate unit, integration, and API tests, with coverage and mutation-testing metrics persisted via Spring Data JPA.

Technologies: Java 17, Spring Boot, Spring Web, Spring Data JPA, WebClient, Maven, REST API, Node.js, ProcessBuilder, Jest, Supertest, Stryker, JUnit 5, Mockito, Git, GitHub

Intelligent Chatbot Development Intern (RAG)

ANP (National Ports Agency), Casablanca | August 2025

- Developed an AI-powered Retrieval-Augmented Generation (RAG) chatbot enabling semantic search across PDF documents.
- Implemented document processing, indexing, and semantic search pipelines using embedding models for accurate information retrieval.

Technologies: Python, Streamlit, LangChain, FAISS, HuggingFace Embeddings, SQLite

Technical Skills

Backend: Java 17, Spring Boot, Spring MVC, Spring Security, Spring Data JPA, Hibernate, Clean Architecture, Spring Modulith, RESTful APIs, Microservices Architecture

Generative AI: Generative AI, LLMs, Retrieval-Augmented Generation (RAG), Prompt Engineering

Databases: SQL, PostgreSQL, MySQL, SQLite, Data Modeling

Testing & Quality: JUnit 5, Mockito, Postman, Jest, Supertest

Frontend: React.js, Vue.js

Tools: Git, GitHub, Maven, Docker, Figma, VS Code

Projects

Carpooling Platform – Microservices Architecture

Java 17, Spring Boot, Spring Security, React.js, MySQL, Docker, Git, GitHub, Postman

- Developed a microservices-based carpooling platform with independent services for users, trips, and bookings.
- Designed and implemented secure RESTful APIs using Spring Security for user, trip, and booking management.

Interactive Educational Learning Platform

Java 17, Spring Boot, React, PostgreSQL, WebSocket, Docker, Git, Postman

- Developed a full-stack educational platform using Java Spring Boot and React.js with real-time learning features.
- Implemented live competitions and progress tracking using WebSocket.
- Designed RESTful APIs and managed course and user data.

Certifications

- **Oracle Java Foundations (Badge)** – Oracle *(Java Syntax, OOP Fundamentals)*
- Java Programming – DataCamp
- JavaScript Essentials 1 – Cisco Networking Academy
- Python Programmer Bootcamp – 365 Data Science
- Introduction to Git – DataCamp

Activities

- Active Member – BYTE BRIGADE & ENIAC Clubs
- Co-founder – Multitalents Club
- Participant – Water4Future Hackathon
- Attendee – Technology Events and Conferences

Soft Skills & Languages

Soft Skills: Analytical Thinking, Problem Solving, Autonomy, Teamwork, Adaptability

Languages: French (Fluent), English (Fluent), Arabic (Native)